

How to create a FLEX system (Part 3) ... In the case of 6800FLEX

 flexonsbd.blogspot.com/2018/07/flex6800flex.html

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00410	45 52 52 4F 52 53 00 00	53 59 53 00 00 01 01 01	ERRORS..SYS.....
00420	09 00 09 02 00 04 0C 4F	46 4C 45 58 32 00 00 00	OFLEX2...
00430	53 59 53 00 00 01 0A 04	04 00 19 00 00 04 0C 4F	SYS.....0
00440	50 52 49 4E 54 00 00 00	53 59 53 00 00 04 05 04	PRINT...SYS.....
00450	05 00 01 00 00 04 0C 4F	43 41 54 00 00 00 00 00	OCAT.....
00460	43 4D 44 00 00 04 06 04	08 00 03 00 00 04 0C 4F	CMD.....0
00470	4C 49 53 54 00 00 00 00	43 4D 44 00 00 04 09 05	LIST....CMD.....
00480	01 00 03 00 00 04 0C 4F	43 4F 50 59 00 00 00 00	OCOPY.....
00490	43 4D 44 00 00 05 02 05	06 00 05 00 00 04 0C 4F	CMD.....0
004A0	54 54 59 53 45 54 00 00	43 4D 44 00 00 05 07 05	TTYSET..CMD.....
004B0	08 00 02 00 00 04 0C 4F	52 45 4E 41 4D 45 00 00	ORENAME..
004C0	43 4D 44 00 00 05 09 05	09 00 01 00 00 04 0C 4F	CMD.....0
004D0	44 45 4C 45 54 45 00 00	43 4D 44 00 00 05 0A 06	DELETE..CMD.....
004E0	01 00 02 00 00 04 0C 4F	41 53 4E 00 00 00 00 00	OASN.....
004F0	43 4D 44 00 00 06 02 06	02 00 01 00 00 04 0C 4F	CMD.....0
00500	00 07 00 00 00 00 00 00	00 00 00 00 00 00 00 00	
00510	41 50 50 45 4E 4A 00 00	43 4D 4A 00 00 06 03 06	APPEND CMD

How to create a FLEX system (Part 3)... In the case of 6800FLEX

How to create a FLEX system (Part 3)... In the case of 6800FLEX

On June 27th, I explained how to create a 6809FLEX system, so this time I will explain how to create a FLEX system for the 6800. Some may say, "

Who would use 6800FLEX nowadays?", but I think there are some people who have fond memories of being engrossed in amateur languages such as VTL, TinyBasic, GAMEIII, and TL1 in the early days of microcomputers. I was one of them.

To run those languages, it is certainly convenient to have an OS.

(If you just want to save it to disk and boot and run it, you can save it in MOT format on a Windows PC and use the Monitor's Load and Save commands...)

That being said, I created 6800FLEX out of technical curiosity, wanting to run FLEX on the 6800.

Similar to the 6809 case, extract FLEX2.SYS from the DSK file in the 6800 folder in SWTPCemu and create FLEX2.COR from it. FLEX2.SYS can be found in various DSK files, but we will use the one in Flex2-01.DSK. This DSK also contains FLEX.COR and FLEX.SYS, but these appear to be outdated, so we will not use them and will create it from FLEX2.SYS.

Specific steps

(All numbers below are in hexadecimal notation.)

(1) Obtain FLEX.COR

003F0	00 00 00 00 00 00 00 00	00 00 00 00 00 00 00 00	
00400	00 06 00 00 00 00 00 00	00 00 00 00 00 00 00 00	
00410	45 52 52 4F 52 53 00 00	53 59 53 00 00 01 01 01	ERRORS..SYS.....
00420	09 00 09 02 00 04 0C 4F	46 4C 45 58 32 00 00 00	OFLEX2...
00430	53 59 53 00 00 01 0A 04	04 00 19 00 00 04 0C 4F	SYS.....0
00440	50 52 49 4E 54 00 00 00	53 59 53 00 00 04 05 04	PRINT...SYS.....
00450	05 00 01 00 00 04 0C 4F	43 41 54 00 00 00 00 00	OCAT.....
00460	43 4D 44 00 00 04 06 04	08 00 03 00 00 04 0C 4F	CMD.....0
00470	4C 49 53 54 00 00 00 00	43 4D 44 00 00 04 09 05	LIST....CMD.....
00480	01 00 03 00 00 04 0C 4F	43 4F 50 59 00 00 00 00	OCOPY....
00490	43 4D 44 00 00 05 02 05	06 00 05 00 00 04 0C 4F	CMD.....0
004A0	54 54 59 53 45 54 00 00	43 4D 44 00 00 05 07 05	TTYSET..CMD.....
004B0	08 00 02 00 00 04 0C 4F	52 45 4E 41 4D 45 00 00	ORENAME..
004C0	43 4D 44 00 00 05 09 05	09 00 01 00 00 04 0C 4F	CMD.....0
004D0	44 45 4C 45 54 45 00 00	43 4D 44 00 00 05 0A 06	DELETE..CMD.....
004E0	01 00 02 00 00 04 0C 4F	41 53 4E 00 00 00 00 00	OASN.....
004F0	43 4D 44 00 00 06 02 06	02 00 01 00 00 04 0C 4F	CMD.....0
00500	00 07 00 00 00 00 00 00	00 00 00 00 00 00 00 00	
00510	41 50 50 45 4E 44 00 00	43 4D 44 00 00 06 03 06	APPEND..CMD.....
00520	05 00 03 00 00 04 0C 4F	45 44 49 54 00 00 00 00	CEDIT....
00530	43 4D 44 00 00 06 06 08	09 00 18 00 00 04 0C 4F	CMD.....0
00540	41 53 4D 42 00 00 00 00	43 4D 44 00 00 08 0A 0B	ASMB....CMD.....
00550	04 00 19 00 00 04 0C 4F	42 55 49 4C 44 00 00 00	OBUILD...

Image 1 Directory

If you open FLEX2-01.DSK with a binary editor and look at the directory starting at address 410, you will see that FLEX2.SYS is a sector track with 18 sectors, numbered 01,0A through 04,04.

The address ranges from 1300 to 2BFF, so copy and extract this data and save it as FLEX2.SYS.

```

01300 02 01 00 01 02 AC 00 35 08 18 3A 00 00 04 00 00 .....5.:.....
01310 00 FF 1B 00 00 00 00 00 00 00 00 00 00 00 .....
01320 00 00 00 00 00 00 00 00 00 00 00 00 00 .....
01330 00 00 00 00 00 00 00 01 00 00 00 00 02 AC 49 .....I
01340 01 00 02 AC 4E 63 46 4C 45 58 20 32 2E 30 04 2B ...NcFLEX 2.0.+
01350 2B 2B 04 3F 3F 3F 04 57 48 41 54 3F 04 43 41 4E ++,???.WHAT?.CAN
01360 27 54 20 54 52 41 4E 53 46 45 52 04 4E 4F 54 20 'T TRANSFER.NOT
01370 46 4F 55 4E 44 04 44 49 53 4B 20 45 52 52 4F 52 FOUND.DISK ERROR
01380 20 23 04 44 52 49 56 45 53 20 4E 4F 54 20 52 45 #.DRIVES NOT RE
01390 41 44 59 04 47 45 54 00 B2 68 4D 4F 4E 00 B3 AB ADY.GET...HMON...
013A0 00 27 10 03 E8 00 64 00 0A 02 AC C0 01 39 02 AC .'.d.....9..
013B0 D8 01 39 02 AC E4 01 39 02 AC F8 C4 01 00 00 00 ..9...9.....
013C0 00 00 00 00 7E AD 57 7E AD 6F 7E AD B8 7E E1 AC .....W.o.....
013D0 7E E1 AC 7E E1 D1 7E E1 D1 7E AF 15 7E AF 5C 7E .....W.o.....
013E0 AE 51 7E AE A7 7E B0 1F 7E AE CF 7E B0 3E 7E AD .Q'.....>..
013F0 F2 7E B0 69 7E B1 FA 7E B1 10 7E B1 E8 7E AF B2 .i'.....~..
01400 02 02 00 02 7E B0 06 7E B2 D9 7E B1 67 7E B0 03 .....8...
01410 7E B1 AC 7E B3 89 7E AE 0D 7E AE 0D 7E AE 0D 8E .....N.....
01420 A0 7F CE AC 4E BD AE A7 BD AE CF 7F AC 11 BD B4 ....N.....
01430 00 7F AC 28 BD B3 FD 8E A0 7F CE AD 03 FF A0 48 ...(.H.....
01440 FF AC 16 CE A7 00 FF A0 12 CE A7 0F FF A0 00 CE .....
01450 AC F8 FF AC 31 7F AC 34 7F AC 4C 8D 5D B8 AC 11 ....1..4..L.]..
01460 B1 AC 02 26 05 7C AC 15 20 16 7D AC 28 27 03 7E ...&.]..).('..
01470 B3 9D BD B4 03 26 B4 CE AC 57 BD AE A7 BD AE 51 .....&...W....Q

```

Image 2: Beginning of FLEX2.SYS

```

328F0 00 00 00 00 00 00 00 00 00 00 00 00 00 .....
32900 04 03 00 17 02 BE 80 23 7E BE B3 7E BF 10 7E BF .....#.....
32910 42 7E BF 59 7E BF 75 7E BF A7 7E BF A7 00 00 00 B'.Y'.u.....
32920 00 00 00 00 00 00 00 00 00 00 02 BE B3 C4 8D .....
32930 3E 86 8C 7D AC 34 27 01 3F 01 0F B7 80 18 BD BF >.]..4'?......
32940 A0 C6 00 B6 80 18 85 02 26 07 85 01 26 F5 16 20 .....&...&..
32950 0B B6 80 1B A7 00 08 5A 26 E9 8D 05 C5 1C 01 0E .....Z&.....
32960 39 7D AC 34 27 01 3F F6 80 18 C5 01 26 F3 39 B1 9].4'?.&..9..
32970 80 19 27 12 B7 80 1B BD BF A0 86 1B B7 80 18 BD ..7..3.....
32980 EF A0 37 8D DC 33 F7 80 1A 7E BF A0 8D E1 86 AC ..7..3.....
32990 7D AC 34 27 01 3F 01 0F B7 80 18 BD BF A0 C6 00 }.4'?......
329A0 B6 80 18 85 02 26 07 85 01 26 F5 16 20 0B A6 00 .....&...&..
329B0 B7 80 1B 08 5A 26 E9 8D A8 C5 5C 01 0E 39 86 8C .....Z&...&..9..
329C0 7D AC 34 27 01 3F 01 0F B7 80 18 BD BF A0 8D 91 }.4'?......
329D0 01 0E C5 18 39 FF BE 97 8D 17 86 0B B7 80 18 8D .....9.....
329E0 3B BD BE E5 FE BE 97 C5 40 26 02 0C 39 C6 0B 0C .....9.....
329F0 39 A6 03 02 BF 77 3A 81 03 23 01 4F 8D 15 F6 80 9...w...#.0...
32A00 04 04 00 18 19 E7 00 B7 80 14 B7 BE 95 8D 08 A6 .....
32A10 00 B7 80 19 0C 20 0D CE BE 99 F6 BE 95 27 04 08 .....
32A20 5A 26 FC 39 BD BF A3 BD BF A6 39 A6 03 81 01 23 Z&..9.....9...#
32A30 C6 C6 8D 0D 39 00 00 00 00 00 00 00 00 00 .....9.....
32A40 00 00 00 00 00 00 00 00 00 00 00 00 00 .....
32A50 00 00 00 00 00 00 00 00 00 00 00 00 00 .....
32A60 00 00 00 00 00 00 00 00 00 00 00 00 00 .....
32A70 00 00 00 00 00 00 00 00 00 00 00 00 00 .....
32A80 00 00 00 00 00 00 00 00 00 00 00 00 00 .....
32A90 00 00 00 00 00 00 00 00 00 00 00 00 00 .....
32AA0 00 00 00 00 00 00 00 00 00 00 00 00 00 .....
32AB0 00 00 00 00 00 00 00 00 00 00 00 00 00 .....
32AC0 00 00 00 00 00 00 00 00 00 00 00 00 00 .....
32AD0 00 00 00 00 00 00 00 00 00 00 00 00 00 .....
32AE0 00 00 00 00 00 00 00 00 00 00 00 00 00 .....
32AF0 00 00 00 00 00 00 00 00 00 00 00 00 00 .....
32B00 00 00 00 19 02 BE A3 10 E1 AC E1 D1 80 04 80 12 .....
32B10 A0 00 A0 12 E0 D0 A0 48 16 AD 00 00 00 00 00 .....H.....
32B20 00 00 00 00 00 00 00 00 00 00 00 00 00 .....
32B30 00 00 00 00 00 00 00 00 00 00 00 00 00 .....

```

Image 3: End of FLEX2.SYS

Looking at the end of FLEX2.SYS, the last three sectors appear to be independent. The two sectors starting at address 2900 are Driver.bin, and the sectors starting at 2B00 are the I/O vectors.

FLEX2.COR is obtained by removing these three sectors and saving up to 282FF.

(2) Prepare driver.bin and io.bin.

I've uploaded the necessary driver.txt and io.txt files to Microsoft OneDrive, so download them and assemble them with the included 6800AsmWin.exe to get driver.bin and io.bin.

Unlike 6809FLEX, you write I/O-related addresses in this driver.bin.

As you can see in driver.txt, after the driver vector starting from BE80, write F000 (INCH), F002 (OUTCH), and F838 (MONITOR) at the location corresponding to address BEA3, F000 (INCH), F002 (OUTCH), and BEAF.

(Actually, this alone is not enough; you also need io.txt for overwriting.)

(3) Convert the BIN file to packet format

. Convert the obtained BIN file to packet format. The method is as described last time: prepare an empty DSK file (BLANK.DSK), set it in the drive of FlexDrvWin.exe, and simply drag and drop the bin file onto it. This will automatically convert it to packet format, and sector information will be added to the beginning of each sector.

(3-1) Convert driver.bin

02000	01 02 00 01 02 BE 80 C4 7E BE F0 7E BF 64 7E BE	d.
02010	B7 7E BE B9 7E BE B0 7E BE C5 7E BE C0 7E BE D5	
02020	7E BE D5 7E BE ED FF FF FF FF FF F0 00 F0 02 00	
02030	00 00 00 00 00 00 00 F8 38 00 00 00 00 00 5F	8.....
02040	39 A6 03 5F 39 A6 03 B7 BE B3 5F 0C 39 A6 03 B7	9..9.....9...
02050	BE B3 5F 0C 39 A6 03 B7 BE B3 5F 0C 39 7F BE B3	...9.....9...
02060	86 43 B7 F0 CC 01 86 55 B7 F0 CC B6 F0 C0 B6 F0	.C.....U.....
02070	CC 47 25 F7 39 5F 0C 39 7F BE B4 36 86 52 B0 BF	.G%.9..9...6.R..
02080	D2 8D 62 B6 BE B3 B0 BF D2 8D 5A 32 B0 BF D2 8D	.b.....22....
02090	54 17 B0 BF D2 8D 4E B6 BE B4 B0 BF D2 B0 BF B8	T.....N.....
020A0	81 15 27 D4 81 1B 26 04 C8 10 5D 39 81 06 26 ED	...'&...}9.&..
020B0	B0 BF D2 FF BE B5 FE BE B5 5F 7F BE B4 B0 BF B8	&.....D.
020C0	A7 00 08 8D 20 5A 26 F5 B0 BF B8 B1 02 BF 44 A8	...Z&.....D.
020D0	BE B4 27 07 86 15 B0 BF D2 20 CF 86 06 B0 BF D2	
020E0	B0 BF B8 81 06 26 F9 5F 39 B8 BE B4 B7 BE B4 39	&..9.....9
020F0	7F BE B4 36 86 57 8D 66 8D EF B6 BE B3 8D 5F 8D	...6.W.f.....
02100	00 00 00 02 E8 32 8D 5A 8D E3 17 8D 55 8D DE B6	2.Z.....U...
02110	BE B4 8D 4E 8D 35 81 15 27 DA 81 1B 26 04 C8 10	...N.5...'&...
02120	5D 39 81 06 26 EE FF BE B5 FE BE B5 5F 7F BE B4	}9.&.....
02130	A6 00 08 8D 2D 8D B6 5A 26 F6 B6 BE B4 8D 23 8D	-..Z&.....#.
02140	DA 81 15 27 E4 81 06 26 F8 5F 39 37 C6 55 86 15	...'&..97.U...
02150	B7 F0 CC 01 B6 F0 CC 47 24 FA B6 F0 C0 F7 F0 CC	G\$.....
02160	33 39 37 C6 55 36 86 15 B7 F0 CC 01 B6 F0 CC 47	397.U6.....G
02170	47 24 F9 32 B7 F0 C0 F7 F0 CC 33 39 00 00 00 00	G\$.2.....39...
02180	00 00 00 00 00 00 00 00 00 00 00 00 00 00 00	
02190	00 00 00 00 00 00 00 00 00 00 00 00 00 00 00	
021A0	00 00 00 00 00 00 00 00 00 00 00 00 00 00 00	
021B0	00 00 00 00 00 00 00 00 00 00 00 00 00 00 00	
021C0	00 00 00 00 00 00 00 00 00 00 00 00 00 00 00	
021D0	00 00 00 00 00 00 00 00 00 00 00 00 00 00 00	
021E0	00 00 00 00 00 00 00 00 00 00 00 00 00 00 00	
021F0	00 00 00 00 00 00 00 00 00 00 00 00 00 00 00	
02200	00 00 00 01 02 AD 09 0C 7E F0 00 7E F0 00 7E F0	
02210	02 7E F0 02 16 AD 00 00 00 00 00 00 00 00 00	
02220	00 00 00 00 00 00 00 00 00 00 00 00 00 00 00	
02230	00 00 00 00 00 00 00 00 00 00 00 00 00 00 00	

Image 4: driver.bin and io.bin

Set the value in the Start Adr box of FlexDrvWin.exe to 0xBE80, then drag and drop driver.bin. The execution address can remain at 0x0000, so answer "Yes" and driver.bin will be saved in packet format from the specified address.

Open the saved Blank.bin with a binary editor, and you will see that it is saved in two sectors starting from address 2000. Copy this out and save it as driver_dsk.bin.

(3-2) Converting io.bin

Unlike console.bin in the case of 6809, in the case of 6800, I/O related vectors are not written here but in driver.bin, so this should be sufficient, but it seems that this io.bin is read at the end of startup and is used to further overwrite the corresponding location in memory that has been set with the value in driver.bin.

(I had a lot of trouble figuring this out.)

When putting it into FlexDrvWin.exe, set Start Adr to 0xAD09 and Exec Adr to 0xAD00.

Copy and extract this one sector and save it as io_dsk.bin.

(4) Connect driver_dsk.bin and io_dsk.bin to FLEX.COR

Prepare a new BLANK.DSK and copy FLEX2.COR to 2000-35FF.

Copy driver_dsk.bin to the two sectors after that, and io_dsk.bin to the sector after that.

(5) Rewrite sector information

0001	01 01 00 01	32 AC 00 25 00 18 3A 00 00 04 00 00	5.....	0000	00 00 00 00	00 00 00 00 00 00 00 00 00 00 00 00				
0002	00 FF 7B 06 00 00 00 00 00 00 00 00 00		0001	01 01 00 01	32 AC 00 25 00 18 3A 00 00 04 00 00	5.....	0001	00 00 00 00	00 00 00 00 00 00 00 00 00 00 00 00	
0003	00 00 00 00 00 00 00 00 00 00 00 00 00		0002	00 00 00 00	00 00 00 00 00 00 00 00 00 00 00 00		0002	00 00 00 00	00 00 00 00 00 00 00 00 00 00 00 00	
0004	01 00 02 AC 4E 63 4E 4C 45 58 29 32 2E 38 04 20	FLEX 2.0..	0003	00 00 00 00	00 00 00 00 00 00 00 00 00 00 00 00		0003	00 00 00 00	00 00 00 00 00 00 00 00 00 00 00 00	
0005	28 28 04 3F 3F 3F 04 51 4E 41 54 3F 04 41 41 4E	TYP 0001 CAN	0004	00 00 00 00	00 00 00 00 00 00 00 00 00 00 00 00		0004	00 00 00 00	00 00 00 00 00 00 00 00 00 00 00 00	
0006	27 54 20 54 52 41 4E 53 4E 45 53 04 4F 54 20	TYP 0001 CAN	0005	00 00 00 00	00 00 00 00 00 00 00 00 00 00 00 00		0005	00 00 00 00	00 00 00 00 00 00 00 00 00 00 00 00	
0007	4E 4F 55 4E 44 04 44 48 53 4E 29 45 53 4F 52	FOUND 0001 ERROR	0006	00 00 00 00	00 00 00 00 00 00 00 00 00 00 00 00		0006	00 00 00 00	00 00 00 00 00 00 00 00 00 00 00 00	
0008	20 29 04 44 52 40 56 45 53 29 4E 4F 54 29 52 45	8 DRIVES NOT RE	0007	00 00 00 00	00 00 00 00 00 00 00 00 00 00 00 00		0007	00 00 00 00	00 00 00 00 00 00 00 00 00 00 00 00	
0009	41 44 59 04 47 45 54 06 60 88 40 4F 4E 00 00 40	ADV 0001 0000	0008	00 00 00 00	00 00 00 00 00 00 00 00 00 00 00 00		0008	00 00 00 00	00 00 00 00 00 00 00 00 00 00 00 00	
000A	00 27 10 00 58 00 64 06 04 03 AC 03 01 39 03 AC		0009	00 00 00 00	00 00 00 00 00 00 00 00 00 00 00 00		0009	00 00 00 00	00 00 00 00 00 00 00 00 00 00 00 00	
000B	08 01 39 03 AC 04 01 39 03 AC 04 01 39 03 AC		000A	00 00 00 00	00 00 00 00 00 00 00 00 00 00 00 00		000A	00 00 00 00	00 00 00 00 00 00 00 00 00 00 00 00	
000C	00 00 00 00 7E AD 57 7E AD 57 7E AD 57 7E AD		000B	00 00 00 00	00 00 00 00 00 00 00 00 00 00 00 00		000B	00 00 00 00	00 00 00 00 00 00 00 00 00 00 00 00	
000D	7E 01 AC 7E 01 01 7E 01 01 7E 01 01 7E 01		000C	00 00 00 00	00 00 00 00 00 00 00 00 00 00 00 00		000C	00 00 00 00	00 00 00 00 00 00 00 00 00 00 00 00	
000E	AE 51 7E AE A7 7E 80 1F 7E AE 0F 7E 80 3E 7E AD		000D	00 00 00 00	00 00 00 00 00 00 00 00 00 00 00 00		000D	00 00 00 00	00 00 00 00 00 00 00 00 00 00 00 00	
000F	7E 01 00 00 7E 01 7E 01 7E 01 7E 01 7E 01		000E	00 00 00 00	00 00 00 00 00 00 00 00 00 00 00 00		000E	00 00 00 00	00 00 00 00 00 00 00 00 00 00 00 00	
0010	00 00 00 00 7E AD 57 7E AD 57 7E AD 57 7E AD		000F	00 00 00 00	00 00 00 00 00 00 00 00 00 00 00 00		000F	00 00 00 00	00 00 00 00 00 00 00 00 00 00 00 00	
0011	7E 01 AC 7E 01 01 7E 01 01 7E 01 01 7E 01		0010	00 00 00 00	00 00 00 00 00 00 00 00 00 00 00 00		0010	00 00 00 00	00 00 00 00 00 00 00 00 00 00 00 00	
0012	AD 7F CE AC 4E 00 AE A7 00 AE 0F 7E AC 11 00 04		0011	00 00 00 00	00 00 00 00 00 00 00 00 00 00 00 00		0011	00 00 00 00	00 00 00 00 00 00 00 00 00 00 00 00	
0013	00 7F CE AC 28 00 FD 0E AD 7F CE AC 00 03 FF AD 48		0012	00 00 00 00	00 00 00 00 00 00 00 00 00 00 00 00		0012	00 00 00 00	00 00 00 00 00 00 00 00 00 00 00 00	
0014	FF AC 16 CE A7 00 FF AC 12 CE A7 00 FF AC 12 CE		0013	00 00 00 00	00 00 00 00 00 00 00 00 00 00 00 00		0013	00 00 00 00	00 00 00 00 00 00 00 00 00 00 00 00	
0015	AC F8 FF AC 31 7F AC 54 7F AC 4C 0D 58 06 AC 11		0014	00 00 00 00	00 00 00 00 00 00 00 00 00 00 00 00		0014	00 00 00 00	00 00 00 00 00 00 00 00 00 00 00 00	
0016	81 AC 02 26 05 7C AC 15 38 16 7D AC 28 27 03 7C		0015	00 00 00 00	00 00 00 00 00 00 00 00 00 00 00 00		0015	00 00 00 00	00 00 00 00 00 00 00 00 00 00 00 00	
0017	89 90 8D 84 03 26 84 CE AC 53 8D AE A7 8D AE 51		0016	00 00 00 00	00 00 00 00 00 00 00 00 00 00 00 00		0016	00 00 00 00	00 00 00 00 00 00 00 00 00 00 00 00	
0018	00 8D FA 81 02 AD 0C 04 00 27 03 CE AD 41 7C AC		0017	00 00 00 00	00 00 00 00 00 00 00 00 00 00 00 00		0017	00 00 00 00	00 00 00 00 00 00 00 00 00 00 00 00	
0019	00 8D 80 88 25 17 CE AC 8C 8D 3F 27 08 FE AC 12		0018	00 00 00 00	00 00 00 00 00 00 00 00 00 00 00 00		0018	00 00 00 00	00 00 00 00 00 00 00 00 00 00 00 00	
001A	27 08 8D 3E 26 04 CE 01 6E 0D 02 89 CE AC 5F		0019	00 00 00 00	00 00 00 00 00 00 00 00 00 00 00 00		0019	00 00 00 00	00 00 00 00 00 00 00 00 00 00 00 00	
001B	86 15 87 AC 29 0D AE A7 7F AC 11 7E AD 6F AC AD		001A	00 00 00 00	00 00 00 00 00 00 00 00 00 00 00 00		001A	00 00 00 00	00 00 00 00 00 00 00 00 00 00 00 00	
001C	13 FF AD 16 FE AD 0D FF AD 04 7F AC 29 7F AC 22		001B	00 00 00 00	00 00 00 00 00 00 00 00 00 00 00 00		001B	00 00 00 00	00 00 00 00 00 00 00 00 00 00 00 00	
001D	7F AC 21 7F AC 26 7F AC 24 39 FF AC 3D CE AD 44		001C	00 00 00 00	00 00 00 00 00 00 00 00 00 00 00 00		001C	00 00 00 00	00 00 00 00 00 00 00 00 00 00 00 00	
001E	AD 00 81 5F 23 02 8D 26 FF AC 41 FE AC 3D AD 00		001D	00 00 00 00	00 00 00 00 00 00 00 00 00 00 00 00		001D	00 00 00 00	00 00 00 00 00 00 00 00 00 00 00 00	
001F	28 18 08 FF AC 3D 4D 36 28 0D FE AC 41 AD 01 FE		001E	00 00 00 00	00 00 00 00 00 00 00 00 00 00 00 00		001E	00 00 00 00	00 00 00 00 00 00 00 00 00 00 00 00	

Image 5: The beginning (left) and end (right) of FLEX2.SYS.

Now that we have all the sectors, we will organize the first 4 bytes of sector information for each sector.

The first 4 bytes of each sector starting from address 2000 are:

I will rewrite it as follows.

(6) Rewrite the SIR (System Information Record) information for sectors 00,03

The unused area starting sector from offset 1D in sectors 00,03 (address 200) is rewritten to 01,1A, and the remaining sector count from offset 21, 09,E0, is rewritten to 09,C7.

		未使用セクタ数		未使用セクタ の先頭 Trk,Sct	
00200	00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00		00200	00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00	
00210	46 4C 45 58 36 38 53 59 53 30 31 00 00 01 1A 4F		00210	00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00	
00220	20 09 C7 0B 0E 0F 4F 20 00 00 00 00 00 00 00 00		00220	00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00	
00230	00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00		00230	00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00	
00240	00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00		00240	00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00	
00250	00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00		00250	00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00	
00260	00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00		00260	00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00	
00270	00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00		00270	00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00	
00280	00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00		00280	00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00	
00290	00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00		00290	00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00	
002A0	00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00		002A0	00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00	

Image 6: SIR information

(7) Rewrite directory information in sectors 00,05

The 24 bytes from offset 10 in sectors 00,05 (address 400) are written as follows: file name (8 bytes), extension (3 bytes), attribute (1 byte, usually 00), unused (1 byte, 00), start Trk,Sct (2 bytes, 01,01), end Trk,Sct (2 bytes, 01,19), size (2 bytes, 00,19), 2 bytes (00,00), date (3 bytes, month, day, year in binary).

	ファイルサイズ	日付	開始Trk, Sct	終了Trk, Sct	
00400	00 06 00 00 00 00 00 00	00 00 00 00 00 00 00 00	00 00 00 00 00 00 00 00	00 00 00 00 00 00 00 00	
00410	46 4C 45 58 32 00 00 00	53 59 53 00 00 00 00 00	01 01 01 01 01 01 01 01	01 01 01 01 01 01 01 01	FLEX2...SYS.....
00420	19 00 19 00 00 00 00 00	07 05 12 00 00 00 00 00	00 00 00 00 00 00 00 00	00 00 00 00 00 00 00 00	
00430	00 00 00 00 00 00 00 00	00 00 00 00 00 00 00 00	00 00 00 00 00 00 00 00	00 00 00 00 00 00 00 00	
00440	00 00 00 00 00 00 00 00	00 00 00 00 00 00 00 00	00 00 00 00 00 00 00 00	00 00 00 00 00 00 00 00	
00450	00 00 00 00 00 00 00 00	00 00 00 00 00 00 00 00	00 00 00 00 00 00 00 00	00 00 00 00 00 00 00 00	
00460	00 00 00 00 00 00 00 00	00 00 00 00 00 00 00 00	00 00 00 00 00 00 00 00	00 00 00 00 00 00 00 00	
00470	00 00 00 00 00 00 00 00	00 00 00 00 00 00 00 00	00 00 00 00 00 00 00 00	00 00 00 00 00 00 00 00	
00480	00 00 00 00 00 00 00 00	00 00 00 00 00 00 00 00	00 00 00 00 00 00 00 00	00 00 00 00 00 00 00 00	
00490	00 00 00 00 00 00 00 00	00 00 00 00 00 00 00 00	00 00 00 00 00 00 00 00	00 00 00 00 00 00 00 00	
004A0	00 00 00 00 00 00 00 00	00 00 00 00 00 00 00 00	00 00 00 00 00 00 00 00	00 00 00 00 00 00 00 00	

Image 7 Directory

(8) Finally

The base of the system disk is now complete. All that remains is to find the DSK file containing the necessary CMD files from SWTPCemu, set it to the drive of FlexDrvWin.exe, and copy it to the system disk.

For example, if the DSK containing COPY.CMD is set to drive 2 and the DSK containing the CMD files you want to copy is set to drive 1, then

2.COPY (sp) 1 (sp) 0 (sp) .CMD (sp is a space)

will copy all CMD files to drive 0.

For details, please refer to the documentation in the TSC folder within the Documentation folder.

(9) Other

- The first two sectors of the system disk, 00,01 and 00,02, must contain the BootLoader. The FLEX command "X" during monitoring for the 6800 has the function of reading these two sectors, setting them to A100, and transferring execution. (The Trk,Sct numbers of the first sector of FLEX2.SYS are written at offsets 05 and 06 of this BootLoader. The FLEX LINK command searches for the SYS file and writes its location here. This ensures that it can boot correctly no matter where the SYS file is located on the DSK.)

Changing the value of AC07 from 00 to 08 will erase the deleted characters when the BS key is pressed, making it easier to use. To do this, change the offset 0F (address 200F) from the beginning of FLEX2.COR to 08. Upload

the BLANK.DSK containing the BootLoader, driver.txt, io.txt, and 6800AsmWin.exe, which are necessary for the above work, to [Microsoft OneDrive with the filename MakeFLEX2.](#)